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Assessment of competitive release of endangered beach mouse (*Peromyscus polionotus ammobates*)

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Motivated by conservation concerns, we assessed whether *Sigmodon hispidus* (hispid cotton rat) affects the realized niche of *Peromyscus polionotus ammobates* (Alabama beach mouse). After experimentally removing *S. hispidus* and accounting for spatial autocorrelation in our data, we found weak evidence for an increase in habitat use by *P. p. ammobates* at locations previously occupied by *S. hispidus*. This pattern is consistent with a niche-constriction hypothesis, and was not observed on 2 control grids where removals were not conducted. Our results suggest that removal of *S. hispidus* from habitat remnants immediately following catastrophic hurricanes ameliorates extinction risk of *P. p. ammobates*, but we also stress the inferential limitations of our data and the need for further investigation of competitive interactions between these species.

Key words: Bayesian analysis, competition, conditional autoregressive model, habitat use, hurricane, *Peromyscus*, realized niche, *Sigmodon*

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Peromyscus polionotus ammobates (the Alabama beach mouse) is an endangered (United States Endangered Species Act) subspecies that currently inhabits approximately 25 km of beachfront dunes on the northern coast of the Gulf of Mexico. This area is susceptible to dramatic destruction from hurricanes. Although permanent habitat loss and fragmentation due to commercial and residential development are the ultimate threat to *P. p. ammobates*, there also may be multiple, synergistically interacting factors that play large roles in determining the proximal risk of extinction (Brook et al. 2008). For example, interspecific interactions may become intensified when hurricanes destroy nearshore dunes frequently used by *P. polionotus* and force surviving mice into inland remnant habitat patches that also are occupied by *Sigmodon hispidus* (hispid cotton rat). Thus, the possibility that *S. hispidus* competes with *P. p. ammobates* is a significant conservation concern because *S. hispidus* is possibly an important proximate factor in a chain of causality that could end in extinction.

Competition between *P. p. ammobates* and *S. hispidus* has not been previously explored. The significant difference in body mass, temporal activity patterns, and diet between the 2 species suggests little potential for competitive interaction. Experimental removal of *S. hispidus* does not significantly affect the demography (Cameron 1977) or resource utilization (Kincaid and Cameron 1982) of *Reithrodontomys fulvescens* (fulvous harvest mouse). Similarly, Turner and Grant (1987) found that neither *R. fulvescens* nor *Baiomys taylori* (pygmy

mice) used a broader range of habitats once *S. hispidus* was experimentally removed.

However, *S. hispidus* has been shown to compete with *Microtus ochrogaster* (prairie vole—Terman 1974; Glass and Slade 1980; Swihart and Slade 1990), and is thought to strongly influence rodent community structure, including the abundance of *Reithrodontomys megalotis* (western harvest mouse) and *Peromyscus maniculatus* (deer mouse—Brady and Slade 2001). Compared to most other rodents, *S. hispidus* has a robust phenotype that may assist its recent northward range expansion (Jones 1960). If *S. hispidus* recently arrived on the Fort Morgan Peninsula of Alabama, *P. p. ammobates* may not yet have evolved a sufficiently differentiated niche from *S. hispidus*.

Normally, large-scale, long-term manipulations of both potentially competing species and their resources are the most satisfactory way of assessing the effects of interspecific competition on population dynamics. However, these sorts of experiments are not always feasible (for ethical, legal, or logistical reasons) when one of the species is endangered, or where answers are needed quickly. The behavior of individuals can respond immediately to changes in abundance of a potential competitor. Behavior can, therefore, be used as a



relatively easily measured indicator of interspecific competition. A behavioral response to competition includes a reduction in use of habitats that are part of the individual's fundamental niche. For example, if one species restricts the niche of another, then species removal should allow the remaining species to seek resources in areas where it previously would not go because of poor foraging returns (exploitation competition) or threat of aggression (interference competition). If competition is occurring, then removal of *S. hispidus* should result in an increase in habitat use by *P. p. ammobates* at locations previously used by *S. hispidus*. We anticipate an increase in capture frequency of *P. p. ammobates* at microhabitats used by *S. hispidus* upon removal of *S. hispidus*. Over the same time period, if *S. hispidus* is not removed, then we should not observe an increase in capture frequency of *P. p. ammobates* at microhabitats used by *S. hispidus*.

Even though behavioral responses to competitive release are more easily measured than numerical responses, demonstrating competitive interactions among mammals is, nonetheless, difficult. In particular, appropriate before–after control–impact experimental designs are notoriously difficult to sufficiently replicate. Here, we collect data at a scale commensurate with the anticipated (fine-scale) response of individuals, and deal with the resulting spatial autocorrelation in our data with a Bayesian conditional autoregressive model to determine if *S. hispidus* is limiting access by *P. p. ammobates* to resources in the form of microhabitats.

MATERIALS AND METHODS

Experimental design.—A preliminary survey conducted with Sherman live traps (H. B. Sherman Traps, Inc., Tallahassee, Florida) in high-quality “source” (Pulliam 1988) habitat in the Fort Morgan Unit of the Bon Secour National Wildlife Refuge (Alabama) suggested that *P. p. ammobates* and *S. hispidus* are partially segregated at a scale of tens of meters. This survey area is predominantly flat and treeless. The most common plants include *Spartina patens*, *Schizachyrium scoparium* var. *littorale*, *Paspalum* spp., *Ipomoea imperati*, and *Andropogon* spp. The well-drained, sandy soil allows relatively small changes in elevation to have large effects on the density and composition of the plant community. Over a distance of 20 m, the elevation may change by approximately 0.5 m. The corresponding changes in plant density and composition provide a template of microhabitat heterogeneity upon which spatial differences in habitat use can be readily measured. Although many of these microhabitats may be useful to one species but not the other, our a priori expectation is that some microhabitats within a grid are useful to both species. Without experimental manipulation, it is difficult to determine if this fine scale of spatial segregation between *P. p. ammobates* and *S. hispidus* is caused by differences in habitat preference or if it is the result of competitive exclusion.

Classic experimental designs often consist of a treatment (e.g., fertilizer) applied over an entire plot where measurements within that plot are analyzed as spatially independent

subsamples. In this scenario, the number of plots is the sample size used to compute degrees of freedom for inferences at a between-plot scale. In our system, we emphasize that individuals should respond at the microhabitat scale to experimental manipulation of a potential competitor. The decision to make a few large-scale manipulations or many smaller-scale manipulations does not change the number of responses measured; it only affects the amount of spatial autocorrelation in the responses. Multiple small and spatially dispersed species removals at the scale of microhabitat homogeneity (e.g., 100 m²) would be extremely difficult to achieve, but would allow treatments to be spaced relatively far apart. Practical constraints forced us to apply species removal manipulations at a relatively large scale (e.g., 1,025 m²). Indeed, we view this treatment area as a systematically aggregated collection of small, random patches. However, unlike microhabitat patches drawn randomly from a large landscape, microhabitat patches within this treatment area are in close spatial proximity and are therefore not spatially independent.

Data collection.—To measure microhabitat use, we established 4 livetrapping grids. The grids were arranged sequentially along an east–west axis approximately 250 m inland from the Gulf of Mexico shoreline, with an intergrid spacing of 150 m (Universal Transverse Mercator of centroid of area: 403152, 3344390). Each grid contained a square, 8 × 8 arrangement of trapping stations, wherein lateral neighbor stations were separated by 15 m. At each trapping station, we deployed 2 live traps (model LFG, 7.2 × 8.9 × 22.9 cm, and model SNA, 5.1 × 6.4 × 15.2 cm; H. B. Sherman Traps, Inc.) separated by 1 m. All traps were set in the late afternoon and then checked beginning around midnight. Because *S. hispidus* is crepuscular and *P. p. ammobates* is nocturnal, the former could have occupied traps before the latter, thereby preventing us from detecting *P. p. ammobates* at a trap station even if it was present. Thus, we made the smaller trap at each station impermeable to *S. hispidus* by covering its entrance with a sheet of aluminum that had a 1.9-cm-diameter hole. This allowed passage of *P. p. ammobates* while excluding *S. hispidus*, thereby enabling us to detect both species at each station. Although individual capture probabilities are not known, the consistency of our trapping protocol should provide an opportunity to detect change in a trapping station's capture probability.

Field protocols followed guidelines of the American Society of Mammalogists (Sikes et al. 2011) and were approved by Iowa State University's Animal Care and Use Committee (IACUC 10-05-6008-W). Between 19 February 2007 and 2 April 2007, we checked all traps once per day, 5 days a week, on biweekly intervals. All traps were checked 16 times during this period, the Preremoval Period. The experimental Removal Period began on 2 April 2007 and continued until 21 May 2007, during which time all *S. hispidus* captured on 2 of the grids were removed (transported ~25 km and released in the Perdue Unit of the Bon Secour National Wildlife Refuge). All traps were checked 15 times after the initiation of removal of *S.*

hispidus. All capture data for *P. p. ammobates* from the first 2 consecutive days immediately after the initiation of removal of *S. hispidus* were excluded from all subsequent analyses because this period was considered transitional between the Preremoval Period and the Removal Period. The decision to exclude these data was made before conducting any inferential data analysis, and was based on our observation of a sharp, asymptotic decline in captures of *S. hispidus* over this period. On the 2 removal grids, the mean number of captures of *S. hispidus* per day after this transitional period was 22% and 28%, respectively, of the mean captures per day prior to removal.

The experiment concluded on 21 May 2007 because fire ants (*Solenopsis invicta*) jeopardized the health of captured animals. Fire ant activity gradually increased on the easternmost grid (a Removal grid) to a level that resulted in unacceptable risks to captured animals (as per our United States Fish and Wildlife Service Endangered Species Permit). Because the data that were collected on this grid after removals of *S. hispidus* began were likely to be significantly affected by the activities of fire ants (Holtcamp et al. 1997), we only used data from the Preremoval Period on this grid. This decision was made prior to inspection of the data.

Data analysis.—We constructed an empirical semivariogram to assess the magnitude of spatial autocorrelation in our trapping data. We 1st divided our data into 7 data sets: 4 Preremoval data sets come from each of the 4 grids before removal of *S. hispidus*, and the other 3 experimental Removal data sets come from the 3 grids after removal that were not affected by fire ants. For all 7 data sets, we then summed the number of times that *P. p. ammobates* was detected at each trap station. Many of these counts are zero because *P. p. ammobates* was never observed at many trap stations. This renders the data significantly nonnormal (skewness = 1.62). Therefore, we applied a Box–Cox transformation to the counts, in which the power parameter (λ) was found with the unconstrained nonlinear optimization function, *fminsearch*, which is available in MATLAB (MathWorks 2009). A spatial semivariance matrix, $\gamma(h) = 0.5(z_x - z_{x+h})^2$, was then computed separately for all 7 transformed data sets. The semivariances at different distances, $\gamma(h)$, were pooled across the 7 data sets. The result is a single semivariogram that reveals the range of spatial scales over which spatial autocorrelation in counts of *P. p. ammobates* occurs. Such autocorrelation is the product of the species' natural history traits within this environment. Pooling $\gamma(h)$ across grids and periods reflects our assumption about the stationarity of the autocorrelation structure, and was necessary to overcome sampling error that would have dominated grid- and period-specific semivariograms.

Because we are interested in the change in occurrence of *P. p. ammobates* at locations where at least 1 *S. hispidus* was observed prior to the initiation of removals, we discarded all data from trap stations where *S. hispidus* was not observed prior to the Removal Period. We modeled the resulting counts (y) of *P. p. ammobates* as realizations of binomial random variables with spatial autocorrelation in the “success”

probability (p). The number of “trials” (N) is the number of times traps were checked for *S. hispidus* during the Preremoval and Removal Periods, which is 16 and 11, respectively:

$$y_{i,j,t} \sim \text{binomial}(N_t, p_{i,j,t}),$$

where i is the trap station in grid j at time period t = “before” or “after” removal of *S. hispidus*.

We model capture probabilities as:

$$g(p_{i,j,t}) = \mu_{j,t} + Z_{i,j,t},$$

using a logit link function, g , with an intercept, $\mu_{j,t}$, common to all trap stations in the same grid at the same time. $Z_{i,j,t}$ is a spatially structured Gaussian random process modeled with a conditional autoregressive structure (Besag et al. 1991; Cressie 1993; Best et al. 1999; Thogmartin et al. 2004). This model can be written as:

$$\log\left(\frac{p_{i,j,t}}{1 - p_{i,j,t}}\right) \middle| \log\left(\frac{p_{(-i,j,t)}}{1 - p_{(-i,j,t)}}\right) \sim \text{normal}\left\{\mu_{j,t} + \sum_{k=1}^8 w_{i,j,k,t} \left[\log\left(\frac{p_{j,k,t}}{1 - p_{j,k,t}}\right) - \mu_{j,k,t}\right], \tau_{i,j,t}^2\right\},$$

where $p_{(-i,j,t)} = \{p_{k,j,t} : k \neq i\}$ for all $k_{\max} = 8$ trap station in the “queen’s adjacency” neighborhood surrounding focal trap station i . Anticipating that the semivariogram analysis would show spatial autocorrelation at the smallest intertrap intervals (15 m), we ameliorated its effects (and potential for spatial pseudoreplication, sensu Hurlbert [1984]) by using the conditional autoregressive model to explicitly model the spatial dependence of counts at a focal trap station on the counts observed at surrounding trap stations. The conditional autoregressive model was fitted with Markov chain Monte Carlo methods, using the GeoBugs function *car.normal* (Thomas et al. 2004). A trap station adjacency matrix was produced with the *adjacency tool* in GeoBugs, and all neighbor weights (w) were set to 1 to produce a standard conditional autoregressive model (Besag et al. 1991). The precision (inverse of variance; τ^2) of the prior distribution on the Gaussian conditional autoregressive was modeled with gamma (0.5, 0.0005), as recommended by Kelsall and Wakefield (1999). A noninformative, flat prior (in a Bayesian analysis, this indicates an a priori equiprobable belief in potential values) was assigned to μ for both time periods (t) and all grids (j). The result of this analysis is a probability density (logit scale) of the mean capture frequency of *P. p. ammobates* during the Preremoval and Removal Periods on 3 trapping grids at trap stations where at least 1 *S. hispidus* was observed before removals began.

RESULTS

Across all 3 grids, we captured 62 unique individuals of *P. p. ammobates* a total of 600 times, and 189 unique *S. hispidus* were captured 1,140 times. Each grid contained trap stations that never captured either species, frequently captured both species, and frequently captured 1 species but infrequently

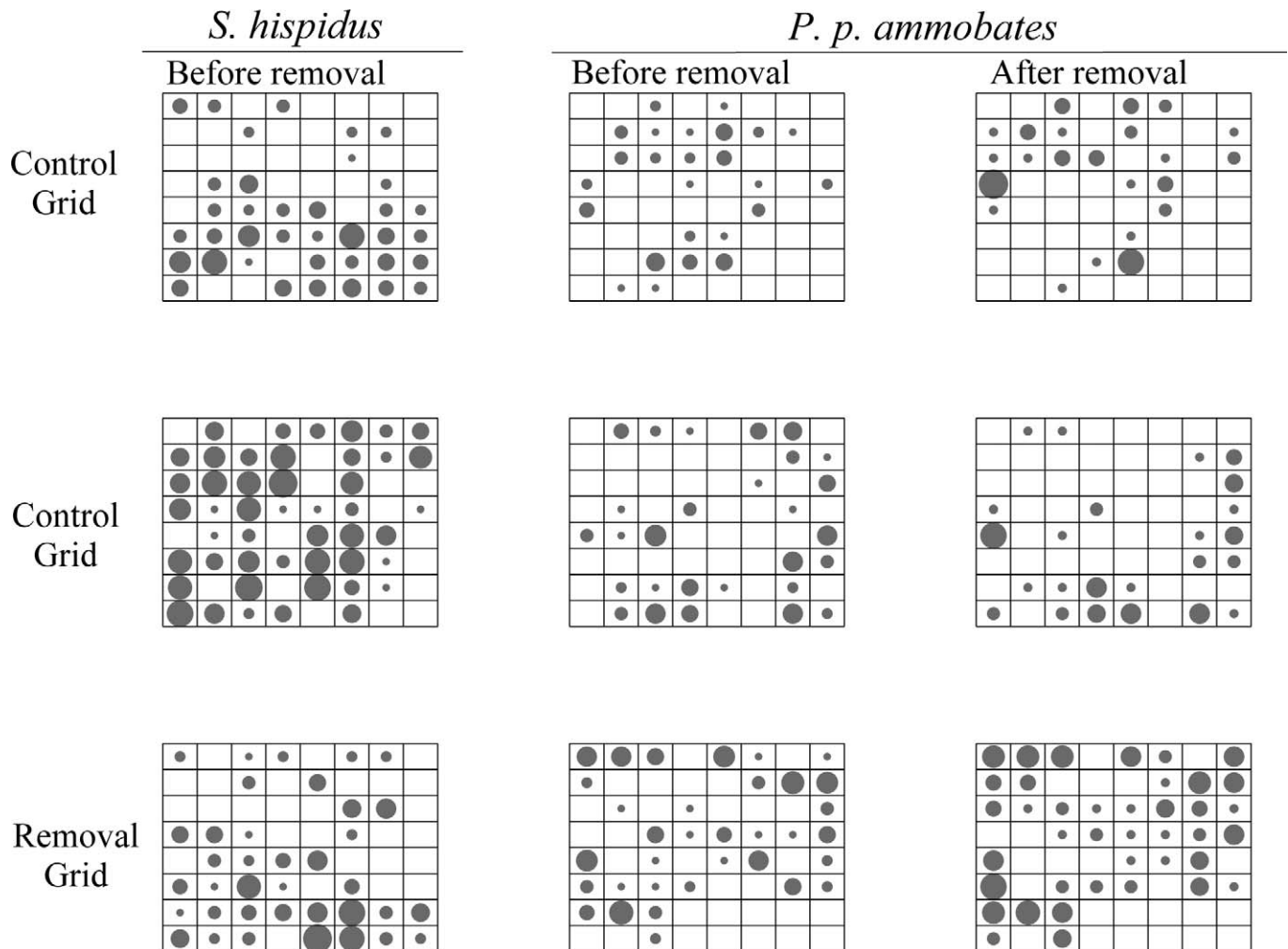


FIG. 1.—Location of captures of *Sigmodon hispidus* (left column) and *Peromyscus polionotus ammobates* (middle and right column) on 3 different 8×8 grids (rows). The sizes of circles are proportional to capture probability. Cells without circles represent trap stations that never captured an animal.

captured the other (Fig. 1). Spatial autocorrelation of captures of *P. p. ammobates* (Fig. 2) was higher (smaller semivariance) for lateral-neighbor traps (15 m) than diagonal neighbors (21.2 m). Spatial autocorrelation of captures of *P. p. ammobates* dissipated after this immediate neighborhood. Thus, the scale of the conditional autoregressive model, which includes only the nearest 8 traps constituting an immediate neighborhood, is commensurate with the observed patterns of spatial autocorrelation.

The Bayesian posterior distributions of the intercepts of the logistic regression of mean detection probability (μ_i) of *P. p. ammobates* at locations where at least 1 *S. hispidus* had been captured prior to the initiation of removals are given in Fig. 3. An increase of μ signifies an increase in habitat use by *P. p. ammobates* at microhabitats used by *S. hispidus*. On one of the control grids where *S. hispidus* was never removed, there is little difference in the mean capture probability of *P. p. ammobates* between the Preremoval Period and the Removal Period (Fig. 3a). On the other control grid, the mean capture

probability of *P. p. ammobates* was higher during the Preremoval Period than the Removal Period (Fig. 3b). Finally, on the treatment grid where *S. hispidus* was removed, the mean capture rate of *P. p. ammobates* was greater after removal than before removal (Fig. 3c). The pattern observed on the treatment grid is qualitatively different from the pattern observed on the control grids.

DISCUSSION

Although each trapping grid is just $1,025 \text{ m}^2$, some trap stations within a grid were likely to contain one species but not the other. We, therefore, looked for an increase in capture probability (i.e., microhabitat use) of *P. p. ammobates* only in places that were used by *S. hispidus* during the Preremoval Period. Our decision to use a single capture of *S. hispidus* as the criterion for including a trap station in our statistical analysis may not have resulted in an optimal subset of the data, because just 1 capture of *S. hispidus* may not be enough use to

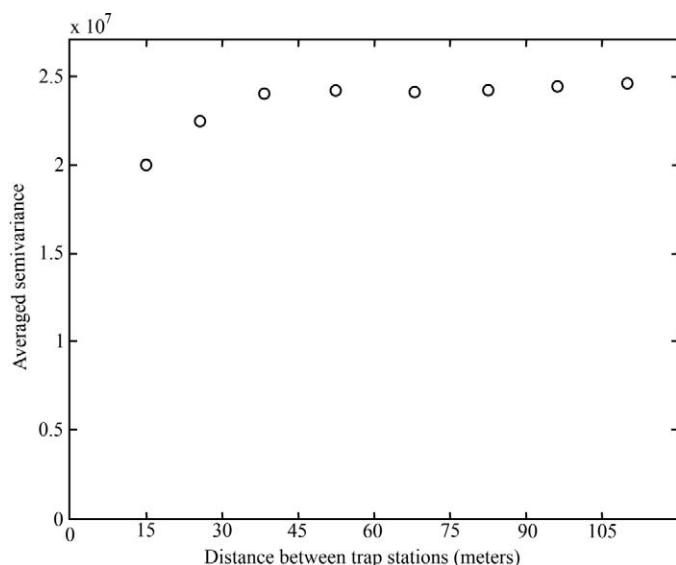


FIG. 2.—Semivariogram of Box–Cox–transformed capture probabilities of *Peromyscus polionotus ammobates*. Capture probabilities were pooled across grids and “before” and “after” removal of *Sigmodon hispidus*.

cause avoidance by *P. p. ammobates*. At the other extreme, including trap stations that frequently recorded *S. hispidus* may have obscured our ability to observe the effects of competition because these locations could have been poor habitat for *P. p. ammobates*, and so we should not expect an increase in capture probability even if significant competitive exclusion occurs.

The change in capture probability of *P. p. ammobates* across time periods is qualitatively different on all 3 grids (Fig. 3). There are 2 extreme interpretations of this observation. The 1st interpretation focuses on grids as experimental units and views the difference between the 2 grids where removals were not performed as an estimate of intergrid variability. Viewed this way, the increase in mean capture probability of *P. p. ammobates* on the removal grid (Fig. 3c) might be attributed to intergrid variation without needing to invoke competitive release. The 2nd interpretation focuses on microhabitats within the grids as the relevant experimental units. Viewed in this way, the difference between the Preremoval and Removal Periods on the 2 grids where removals were not performed (Figs. 3a and 3b) can be averaged together to estimate effect of time (i.e., seasonality) on the capture probability of *P. p. ammobates*. This implies a moderate decline in mean capture probability of *P. p. ammobates* due to environmental change. The increase in mean capture probability of *P. p. ammobates* on the removal grid (Fig. 3c) can then be attributed to release from competition that overwhelms the background seasonal decline in capture probability. A 3rd interpretation of the patterns in Fig. 3 invokes both of the foregoing interpretations.

If the attributes of microhabitats where at least 1 *S. hispidus* was observed vary across grids, and the intensity of competitive interactions covary with these (unmeasured)

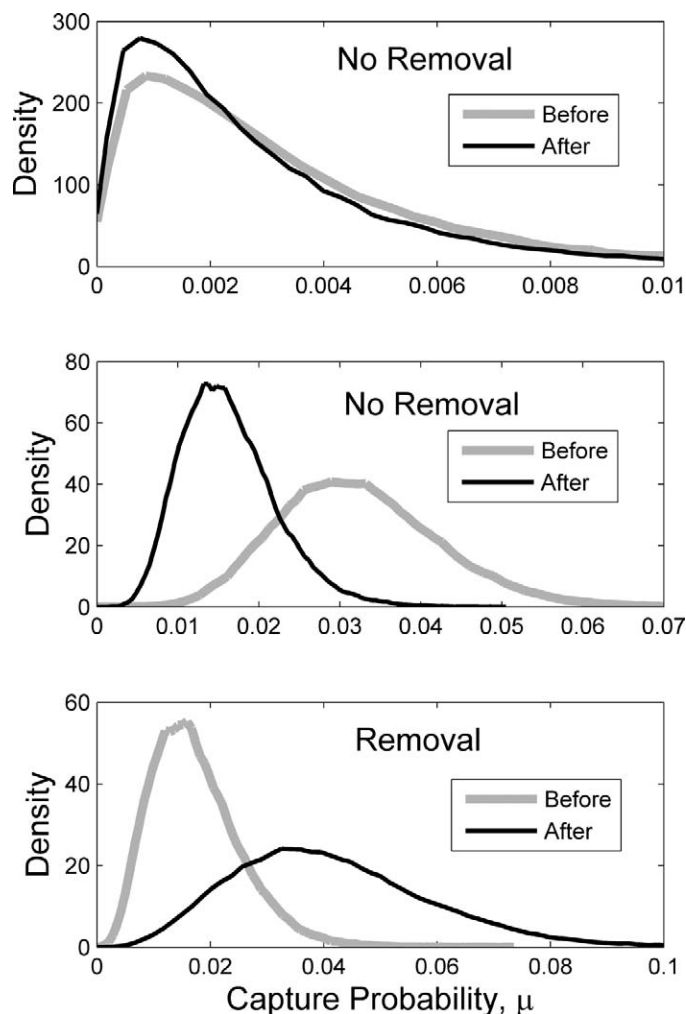


FIG. 3.—Bayesian posterior probability of mass of intercept (μ) of the probability of capture of *Peromyscus polionotus ammobates* at trap stations where at least 1 *Sigmodon hispidus* had been captured prior to the initiation of removals.

attributes, then we should expect intergrid differences in our ability to detect competitive release. We also believe that temporal changes in capture probability occurred during this study. Indeed, foraging behavior of *P. p. ammobates* could change daily in response to environmental cues, and longer-term changes in habitat use can covary with phenology. Furthermore, there could be an interaction between space and time wherein some locations have a greater phenological response than others. Examination of our data cannot address how much variability in capture probability can be attributed to spatial, temporal, and competitive effects, but we believe that all of these effects influenced our observations.

We looked for signs of behavioral response to competitive exclusion at a very small spatial and temporal scale. Studies measuring a response to competitor exclusion at larger spatial scales may not need to deal with spatially autocorrelated data, but the response may take longer to unfold. Indeed, the numerical (i.e., population size) response of mammals with

long generation times may occur over a prohibitively long period. As the time span of a manipulative field study increases, so does the potential for confounding with unmeasured factors, which will result in a larger number of spatial replicates for a given level of statistical power. Furthermore, under crisis conditions with an endangered species, time may be the single biggest limiting factor in making effective management decisions. This method of identifying a competing species is both quick and relatively inexpensive for management organizations with limited budgets and little available labor.

The effect of competition that we observed occurs at a spatial scale smaller than the home-range size of *P. p. ammobates*, which averages 3,586 m² (Swilling and Wooten 2002). Part of an individual's decision to establish a home range may include a joint consideration of the proportion of habitats within a potential home range that provide resources both to itself and *S. hispidus*, and the probability that *S. hispidus* will exploit those resources. However, these considerations can be dramatically altered by the occurrence of catastrophic hurricanes that periodically devastate *P. p. ammobates* and its habitat (Swilling et al. 1998; Pries et al. 2009). "Source" habitat of *P. p. ammobates* (Pulliam 1988) may be eliminated by a hurricane, forcing surviving individuals into "sink" habitats heavily used by *S. hispidus*. Falcý and Danielson (2011) have demonstrated that enhancing sink populations can more effectively promote long-term viability than hastening recovery of damaged source habitat.

Although the robustness of our results was compromised by small sample size and further investigation of the competitive interaction between *P. p. ammobates* and *S. hispidus* is needed, we found evidence that the niche of *P. p. ammobates* is constricted by *S. hispidus*. Therefore, we recommend the removal of the latter from posthurricane habitat remnants to effectively increase the amount of usable habitat for *P. p. ammobates* during periods of crisis for the endangered mouse. Removing *S. hispidus* is both easy and safe to implement: live traps can be set during times when *S. hispidus* is active but *P. p. ammobates* is not, so *S. hispidus* can be captured without significant concern for unintended detention of *P. p. ammobates* and, if deemed necessary, live traps also can be easily modified to allow the selective escape of the smaller *P. p. ammobates* but not the much larger *S. hispidus*.

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